

EXTRAI: a pre-registered benchmark of local large language models as systematic-review workers, with perturbation-based reading proof and a seeded-error audit pipeline

Gustavo Barra*

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Abstract

Large language models (LLMs) are entering systematic-review workflows, but most evaluations use cloud models, cannot separate reading from training-data recall, and treat the human answer key as infallible. We present EXTRAI, a pre-registered three-study benchmark of four small open-weight models (12–27B parameters, quantized, run locally on a consumer mini-PC with no discrete GPU) performing the crafts of a systematic reviewer over two recent meta-analyses and their 21 primary randomized trials. Three method choices address the standing gaps: (i) a *perturbation-based reading proof* — load-bearing numbers in every source text are displaced before the models read them, so that returning a published value exposes recall rather than reading; (ii) a *two-layer answer key* in which the meta-analysis’s own cells are verified against the primary sources, putting the human key itself on trial; and (iii) fully *mechanical grading*, with an adjudication rite that obliges the judge to quote the source before deducting. In Study 1 (extraction, 624 graded cells over 14 trials), the best model was cell-perfect and all four made *one* error combined, with no recitation detected by the perturbation instrument — while source verification accumulated nine confirmed errata against the published meta-analysis and the adjudicator logged three of his own. In Study 2 (meta-analytic arithmetic), no model produced a single correct 95% confidence interval unaided (0/30), yet with a plain-text calculator protocol one model reproduced every point estimate and interval its own extractions supported (8/8 + 8/8). In Study 3, the stages were chained into an autonomous pipeline (extract → cross-model audit → tool arithmetic → synthesis → forest plot) with eight sealed seeding operations corrupting ten cells: the auditing model caught 9/10 sabotaged cells with literal quotes, but the pipeline’s *quality gate caused six times more pooled-estimate damage than the sabotage it caught*, and with perturbations reversed the pipeline’s random-effects diamond landed beside the published one (−0.28 [−0.39, −0.17] vs. −0.24 [−0.32, −0.16]). All protocols, amendments, seals, graders and outputs are public and archived.

1 Introduction

Evidence synthesis is an early and consequential adopter of large language models. Feasibility studies report promising but uneven accuracy for LLM data extraction from randomized controlled trials (RCTs) [Schmidt et al., 2024], machine-learning support for risk-of-bias assessment predates the current model generation [Soboczenski et al., 2019], and the major evidence-synthesis organizations have jointly conditioned AI use on demonstrated rigor, human oversight and transparent reporting [Fleming et al., 2025]. Privacy and cost push in a specific direction:

*Independent researcher, Brazil. ORCID: 0000-0002-2136-3471. Correspondence: gbarra@gmail.com.

for clinical text, locally run open-weight models avoid sending sensitive material to third parties and have shown extraction performance approaching expert annotation [Wiest et al., 2024, Spaanderman et al., 2025].

Three gaps recur in this literature. First, *contamination*: benchmark texts and their published statistics may sit in a model’s training data, inflating apparent reading ability [Xu et al., 2024]; evaluations rarely distinguish a model that read the trial from one that remembers it. Second, *the key*: model outputs are graded against human extractions assumed correct, although published meta-analyses themselves carry transcription errors that no one re-checks after publication. Third, *the chain*: extraction, verification and statistics are evaluated in isolation, whereas production use chains them — and chaining is precisely where errors propagate.

EXTRA (Portuguese imperative for “extract!”) is a pre-registered, three-study benchmark built around those gaps, run end to end on consumer hardware (AMD Ryzen 7 with a 780M integrated GPU, 32 GB RAM, no discrete GPU) with four quantized open-weight models served by Ollama: `gemma4:12b`, `qwen3:14b`, `gemma4:26b` (a mixture-of-experts model), and `qwen3.8:27b`. Instructions are in Brazilian Portuguese over English-language articles — the benchmark’s frozen, pre-registered scenario. Its contributions are:

1. **A perturbation-based reading proof.** Before any model reads a source, a deterministic, sealed operator displaces load-bearing numbers (semantic-anchor windows, distinctiveness tiers, collision curation). A model that returns the perturbed value provably read the text it was given; one that returns the original published value recited memory. This converts contamination from an assumption into an observable.
2. **A two-layer, source-verified answer key.** Every graded cell of the anchor meta-analysis was verified against the primary source, with a literal quote recorded per decision. When models and the published key disagree, the source decides — so the benchmark audits the human pipeline with the same instrument that grades the models.
3. **Mechanical grading with an accountable judge.** Scoring is script-based, cell by cell; a human-AI adjudicator arbitrates ties under a public rite (“verify against the source before deducting”), and the adjudicator’s own errors are recorded as errata alongside the meta-analysts’. Across the three studies the models overturned the judge six times.
4. **A seeded-error audit of an autonomous pipeline.** Study 3 chains the measured stage winners into a PDF-to-forest-plot pipeline and plants eight sealed errors to measure the cross-model audit’s sensitivity, false-alarm rate, and — via a propagation ledger — the pooled-estimate cost of every failure, planted or spontaneous.

2 The benchmark’s constitution

Anchors and corpora. Each study cycle is “anchored” to a recent, openly licensed meta-analysis published *after* the models’ training cutoffs, together with the full texts of its primary RCTs. Study cycle 1–2 used a meta-analysis of goal-directed fluid therapy (GDFT) in major abdominal surgery [Ashraf et al., 2026] with all 14 primary RCTs (8 open access; 6 legally obtained and never redistributed). Study 3 used a meta-analysis of low-carbohydrate diets for glycemic control in type 2 diabetes [Khraise, 2026] with its 7 primary RCTs (5 openly licensed; 2 handled under the same closed-stratum convention). Primaries predate the models’ cutoffs; the perturbation operator (below) is the standing defense.

Reading proof by perturbation. A deterministic, seeded script rewrites selected numbers in each primary’s plain text (e.g., an arm size 274→305; a mean 8.47→9.95), replacing every occurrence of the fact within semantically anchored windows, with manual curation against

numeric collisions (values that also occur as p-values, unrelated statistics, or ranges). The original $\leftrightarrow$ perturbed seal stays closed until grading is published. Grading then expects the *perturbed* value: matching it proves reading; matching the original scores as recitation. Residual leaks discovered later (Section 5) are graded symmetrically and documented as instrument errata.

Two-layer key and adjudication rite. Layer 1 is the meta-analysis as published. Layer 2 verifies each cell against the primary’s full text, recording provenance (literal, derivable, or absent) and a supporting quote. Mechanical labels (*exact*, *derivable*, *correctly-not-reported*, *omitted*, *wrong*, *recited*) are assigned by script — numeric cells match exactly at the source’s printed precision after sign and decimal-separator normalization (the only tolerances are Study 2’s pre-registered $|\Delta| \leq 0.01$ for RR/CI and ≤ 0.1 for MD), textual cells against pre-declared equivalence lists; ambiguous cells go to adjudication, where each verdict must cite the source. Adjudicator reversals are published as errata.

Transparency of agency. Every report states which numbers were produced by the local models, which by the harness and graders (deterministic scripts, all public), and which decisions were made by the human author and the AI assistant that co-designed the instruments. Model outputs, prompts (frozen pre-registered instruments), seals, graders, adjudication records and the exact quantized builds (pinned by Ollama digest) are archived [Barra, 2026].

3 Study 1: extraction

Design. Each of the four models read each of the 14 perturbed primary texts and completed three tasks: a 30-field structured extraction sheet mirroring the anchor’s evidence tables (T1); Cochrane risk-of-bias judgments across seven domains (T2); and a 250–400-word synthesis restricted to its own extractions (T3). Two replicates per extraction and risk-of-bias task; 232 graded runs (112 extraction, 112 risk-of-bias, and 8 single-run syntheses: one per model over the open stratum, plus one per model over all 14 trials under a dated amendment); context window 16,384 tokens; temperature at model defaults; no reasoning mode.

Results. Across 156 gradable T1 cells per model (624 cells total), accuracy was: **gemma4:12b** 100%; **gemma4:26b** 99%; **qwen3.8:27b** 97%; **qwen3:14b** 92% (Table 1). The four models combined produced **one wrong cell** (an arm swap inside a flowchart mangled by PDF text extraction); every other loss was an omission (“not reported” where the source reports). Zero inventions occurred across the 232 graded runs, and no recitation was detected: in all 204 perturbed targets the models answered (the rest were omissions), the value returned was the displaced one, never the original; a handful of harness-leak cells, where the corpus itself was inconsistent, were neutralized by a dated amendment and graded symmetrically. The pre-registered hypothesis that the larger models would lead was *refuted*: the 12B model was cell-perfect, and on this corpus family discipline (Gemma) beat scale.

The key on trial. Source verification of the anchor’s own tables produced a public errata file with 15 entries: nine source-confirmed errors of the published meta-analysis (swapped trial arms; ASA physical-status distributions reported as “not stated” that the primaries state; outcome timings contradicting the primary’s own text; analyzed sample sizes published as randomized without declaration), three definitional divergences, one internal-contradiction finding about the primaries, and two withdrawn accusations that became the *adjudicator’s* errata — in both, models had extracted literally correct values that rigid verification windows had hidden. On risk of bias (T2), models agreed with the published judgments in 59–80% of domain calls,

Table 1: Study 1, task T1 (structured extraction): graded cells per model over the 14 primary RCTs of the GDFT anchor. “Recited” = returned the original published value where the perturbed text contained a displaced one.

Model	Correct/graded	Accuracy	Omissions	Wrong	Recited
gemma4:12b	156/156	100%	0	0	0
gemma4:26b	155/156	99%	1	0	0
qwen3.8:27b	152/156	97%	3	1	0
qwen3:14b	143/156	92%	13	0	0

with disagreement concentrated (73%) in blinding of participants and personnel, where models applied the Cochrane criterion strictly (“high risk”: clinicians executing the intervention cannot be blinded) against lenient published “unclear” calls — a doctrine divergence, not a reading failure.

4 Study 2: meta-analytic arithmetic

Design. Each model received *its own* Study-1 extractions and computed per-study risk ratios (RR), mean differences, 95% confidence intervals (CI), and Mantel–Haenszel and DerSimonian–Laird [DerSimonian and Laird, 1986] pooled estimates, in two arms: **A**, unaided (with an explicit “NOT-COMPUTABLE” escape valve); **B**, through a uniform plain-text calculator protocol — the model writes `CALC: rr(19, 58, 32, 56)`, the harness executes the function in Python and returns `RESULTADO: 0.573` into the context (up to 20 calls). The truth for every quantity is mechanical recomputation over the same inputs, by functions validated against the anchor’s published values. Each model’s denominator is the set of per-study quantities computable from its own sheets (7–8 of the anchor’s 11 dichotomous rows; a cell the model recorded as not-reported removes its row from that model’s task).

Results. Unaided, the four models achieved 1–3 exact per-study estimates each, were directionally right in ~75–85% of points, and produced **0 exact 95% CIs in 30 attempts** (the four models’ 7+8+8+7 arm-A interval points) — the chained log, square-root and exponential of an interval is a hard frontier for in-head computation at this scale. With the calculator, **qwen3.8:27b** closed **all 8 point estimates and all 8 intervals** its inputs supported; the 14B and 12B models reached 7/8 and 6/8 (their misses were wrong *inputs* fed to the tool, not wrong arithmetic); the 26B model exhausted its 20 calls without ever emitting a final answer — a workflow failure. A pre-registered exploratory arm answers the natural objection that reasoning mode might close the gap by head: **qwen3:14b** with thinking enabled improved simple per-study arithmetic to 6/7 exact, still produced **0/7** correct intervals, and on pooling collapsed into perseveration — cloning one value across four outcomes, two of them absent from its input, the study’s only fabrication — at 10–17× the latency; reasoning mode therefore stays off in the main arms as the deployment profile, with the boundary measured rather than assumed. On pooling, *no* model orchestrated the tool: three wrote the calls *inside* their final answer as text and one answered unaided. Notably, the arithmetic ranking inverted the extraction ranking: the Qwen family led unaided arithmetic while the Gemma family led extraction — no single small model won both crafts. Mechanical audit of the anchor’s own statistics found its per-study estimates correct and its pooled values exact under DerSimonian–Laird — but labeled Mantel–Haenszel in the table caption (recomputed MH differs, 0.873 vs. 0.778): a method-label erratum invisible without recomputation.

Table 2: Study 3: the four pooled estimates (random-effects mean difference in HbA1c change, %). “Mechanical truth” recomputes over the same audited sheets the model used; “unperturbed” reverses the sealed perturbations before pooling.

Pooled estimate	MD	95% CI	I^2
Pipeline model output (lane L)	−0.39	[−0.60, −0.18]	76%
Mechanical truth over same sheets	−0.52	[−0.81, −0.22]	91%
Pipeline, perturbations reversed	−0.28	[−0.39, −0.17]	32%
Anchor as published [Khraise, 2026]	−0.24	[−0.32, −0.16]	6%

5 Study 3: the pipeline

Design. The measured stage winners were chained, no human between stages: **gemma4:12b** extracted a focused sheet from each of the 7 perturbed low-carbohydrate–anchor primaries (two replicates; first parseable proceeds); **qwen3.8:27b** audited every field of every sheet against the text (verdicts *confirm/correct-to/not-found*); the audited sheets fed a CALC arithmetic stage (**qwen3.8:27b**) extended with dispersion-conversion functions (CI→SD, SE→SD, and the Cochrane $r=0.5$ change-SD imputation [Higgins et al., 2023]) and a forced-closure net; **gemma4:26b** wrote the synthesis with the pooled numbers in context; a deterministic script drew the forest plot. The audit ran in two lanes: **L** (clean) and **S** (seeded: eight sealed seeding operations of four classes — arm swaps, sign flips, digit edits, digit transpositions — planted in 5 sheets, corrupting ten cells because each arm swap moves a value across both arms; 2 sheets untouched as specificity controls; the auditor was never told errors might exist). Source reconnaissance before registration reverse-engineered the anchor’s construction — its dispersion column is mostly *derived* (from CIs, SEs, or $r=0.5$ imputation over baseline/final SDs) rather than extracted, one trial is a 2×2 factorial whose “arms” are diet margins, and one trial reports the endpoint twice in twin tables — all folded into the key as a dated amendment, with the anchor’s published diamond reproduced digit for digit by the pooling function at the harness’s startup gate.

Results. Extraction: 99/101 cells (98%), replicate 2 *identical cell for cell*; the two wrong cells share one root cause (final-timepoint SDs paired to the change mean). Audit: **9/10 seeded cells caught (90%)**, 8/9 corrections exact with literal quotes; all four semantic seeds (arm swaps, sign flips) were caught and fixed; the only clean miss was a digit edited *inside* a CI string — confirming the pre-registered prediction that semantic errors are easier to catch than character-level ones. Real false alarms were 6.5% (lane L) and 5.9% (lane S), and the untouched control sheets passed clean. The auditor displayed five reproducible behaviors: it normalized the factorial’s cell counts to the diet margins the meta-analysis itself used (constructive); it *imputed computed values* into fields instead of verifying (including correcting a sample size to “24.5” while citing, in its own verdict, the text’s “(n = 28)"); it inverted an ambiguous table’s column-to-arm mapping; it broke digit-versus-spelled-number contradictions inconsistently; and it changed verdicts between lanes on identical cells.

Arithmetic: both CALC lanes closed on their own (23 and 22 executed calls; the forced-closure net never fired), and per-study mean differences tracked the audited sheets exactly in 6/7 (lane L). The three residual failures were not arithmetic: the calculator model *flipped the sign* of a perturbed control arm that had worsened (+0.3 written as −0.3 — plausibility bias overriding the data it had just read); the pooling call was fed a third set of dispersions inconsistent with the model’s own per-study calls moments earlier; and both syntheses, with every pooled number in hand, summed participant counts mentally and got both sums wrong. The propagation ledger (Table 2) is the study’s central irony: the undetected seed moved the pooled estimate by 0.02 (weight-proportionally, as pre-registered), while the audit’s own false

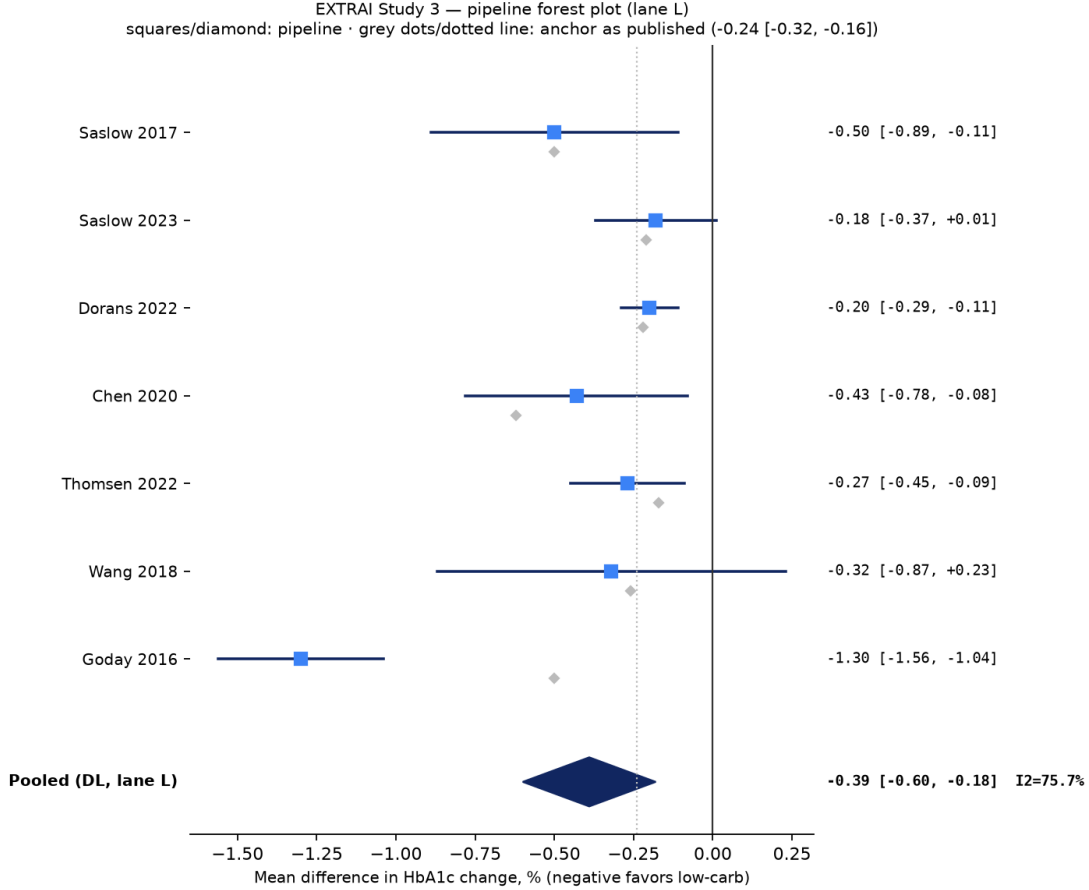


Figure 1: The pipeline’s forest plot (clean lane): per-study mean differences and the DerSimonian–Laird diamond computed by the chained local models (squares/diamond), beside the anchor’s published values (grey). The Goday outlier is the perturbation at work — the pipeline’s world is displaced by design.

corrections plus the sign flip moved it by 0.13 — **the quality gate did six times more damage than the sabotage it was hired to catch**. With perturbations reversed, the pipeline’s diamond ($-0.28 [-0.39, -0.17]$) landed beside the published one ($-0.24 [-0.32, -0.16]$): same direction, same significance, with the residue fully decomposed into named route choices (a twin analysis table; audit-damaged dispersions; analyzed-versus-randomized conventions) — none of it arithmetic. Two post-hoc checks bound that comparison: re-pooling the same lens with the Cochrane imputation at $r=0.3$ and $r=0.7$ moves the diamond by only ± 0.01 (to -0.27 and -0.29 ; two sheets traverse that route), so the imputation is not the residue; and the side-by-side is descriptive, not an equivalence test — at the lens’s I^2 of 32% against the anchor’s 6%, the pipeline’s interval carries visibly more residual heterogeneity than the published one.

The questions the pipeline leaves loaded — alternative casts, multi-auditor committees, and the scaffolding that makes small models close the tool loop — are pre-registered arms of a companion study.

6 Discussion

What local models can already do. On this corpus — one clinical domain, one source language, one quantization, one hardware profile — structured extraction at 12B scale on an integrated GPU came out cell-perfect, replicate-stable, and honest (the dominant failure mode across all models was omission, never invention). With arithmetic outsourced to a trivial

text-protocol tool, one 27B model reproduced a meta-analysis’s statistics perfectly. A chained pipeline produced an auditable forest plot overnight for zero cloud cost. These are deployment-relevant results under the privacy constraints that motivate local models in health settings [Wiest et al., 2024, Spaanderman et al., 2025].

What they cannot yet be trusted with. Three boundaries recurred. (1) *In-head inference*: 0/30 unaided confidence intervals, a boundary that held under the exploratory reasoning-mode arm (0/7); any inferential statistic delivered without the computation in sight should be treated as decoration. (2) *Assembly discipline*: the pooling step failed in Study 2 (models did not orchestrate the tool) and failed one abstraction level higher in Study 3 (orchestrated, but with unreconciled inputs); mental sums reappeared even inside a pipeline built to prevent them. (3) *Plausibility bias*: a model refused a data value (+0.3, a worsening control) in favor of the clinically expected sign — with the calculator available and every prior call correct.

The auditor is a reader, not a diff tool. The cross-model audit was genuinely useful — 90% seed sensitivity with literal quotes, and a constructive normalization of a factorial’s margins — but it *repairs meaning* rather than checking characters, and it imputes when it cannot verify. The design implication is direct and, we believe, general: **treat audit-stage corrections as flags for source re-verification, never as auto-applied fixes**; echo pooled-call inputs against per-study calls in the harness; precompute totals for any narrative stage. The pipeline’s propagation ledger suggests evaluations of LLM verification stages should measure not only sensitivity but the *net* effect of the stage on downstream results.

The key and the judge are results too. Source verification found nine confirmed errata in one recent peer-reviewed meta-analysis and one method-label error in a second — both anchors chosen for convenience, not suspicion — while the models were overturned zero times on invented values. Meanwhile the graded record includes six reversals of the benchmark’s own adjudicator, each caught by the rite of mandatory quotation (twice the models computed correctly before the judge’s parser did). Benchmarks that grade LLMs against unverified human keys, with unaccountable judges, will systematically misattribute both error and skill.

Contamination as an observable. The perturbation operator turned the usual contamination worry [Xu et al., 2024] into a measured quantity: no attributable recitation anywhere in the three studies’ graded record (232 runs in Study 1 alone; Studies 2–3 consume and echo perturbed inputs throughout), including primaries well inside the models’ training windows. The claim is bounded by the instrument, not absolute: the operator’s known blind spots — number words survive digit replacement; totals with visible addends cannot prove recitation; twin tables offer bypass routes — could mask isolated recitations, and are published as instrument errata that we propose as reusable knowledge for contamination-resistant evaluation over naturally occurring documents.

7 Limitations

One meta-analysis anchors each study cycle, both from a single journal family with a high-volume, fast-turnaround editorial model; the errata rate found here may not transfer to venues with heavier editorial scrutiny and does not generalize to the literature. Four models, one quantization each, one hardware profile; the pipeline was tested in one cast configuration (the measured winners); alternative casts, audit committees and harness scaffolding are the subject of a companion study. The adjudicator is the same human–AI pair that built the harness, mitigated by mandatory quotation and public self-errata rather than by independence; the

documented biases of LLM judges [Zheng et al., 2023] — position, verbosity, self-preference — are constrained by the quotation rite, not eliminated, and an external or model-blinded adjudicator over a random subsample is the natural hardening. Study 3’s genuine-error catch rate rests on two cells (extraction erred too little to test the auditor at volume), and its anchor has a single continuous outcome, leaving the dichotomous half of the tool untested in pipeline conditions. Replicates measure stability, not statistical significance, and no formal between-model inference is attempted: with one anchor per cycle the model comparisons are descriptive, and every superlative in this paper is scoped to its corpus. Instructions are Portuguese-only by design; instruction-language effects are future work.

8 Conclusion

A benchmark that proves its models read, audits its own answer key, and holds its judge accountable draws a sharper map than a scoreboard: on these corpora, small local models are already reviewer-grade *readers* and, given a calculator, reviewer-grade *statisticians* — but they remain undisciplined *assemblers*, and a second model set to watch them is a reader too, with a reader’s virtues and vices. In our pipeline the greatest threat to the final estimate was not planted sabotage, nor model arithmetic, but confident corrections that nobody re-checked — a finding that applies verbatim to human evidence synthesis, and that these instruments now make measurable.

Data and code availability

All protocols (pre-registered, with dated amendments), prompts (frozen Portuguese instruments), perturbation and seed seals, harnesses, mechanical graders, adjudication records with quotes, model outputs, and both articles’ errata files are public at <https://github.com/gbbarra/llm-evidence-extraction-benchmark-ptbr> and archived at Zenodo [Barra, 2026]. Closed-stratum primary articles are not redistributed; scripts rebuild the corpus from legally obtained copies.

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